

2. Let  $X = 3 \tan \theta \rightarrow dX = 3 \sec^2 \theta d\theta \therefore \int \frac{dx}{\sqrt{9+x^2}} = \int \frac{3 \sec^2 \theta d\theta}{3 \sec \theta} = \int \sec \theta d\theta = \ln |\sec \theta + \tan \theta| + C = \ln \left| \frac{\sqrt{9+x^2}}{3} + \frac{x}{3} \right| + C$
3. Let  $X = 2 \tan \theta \rightarrow 4+x^2 = 2 \sec^2 \theta, dX = 2 \sec^2 \theta d\theta \therefore \int \frac{dx}{4+x^2} = \frac{1}{2} \int \frac{2 \sec^2 \theta d\theta}{2 \sec^2 \theta} = \frac{1}{2} \int \frac{1}{\sec^2 \theta} d\theta = \frac{1}{2} \int \cos^2 \theta d\theta = \frac{1}{2} \int \frac{1+\cos 2\theta}{2} d\theta = \frac{1}{4} \left( \theta + \frac{\sin 2\theta}{2} \right) + C = \frac{1}{4} \left( \frac{1}{2} \tan^{-1} \frac{x}{2} + \frac{\sin 2 \tan^{-1} \frac{x}{2}}{2} \right) + C$
5. Let  $X = 3 \sin \theta \rightarrow \sqrt{9-X^2} = 3 \cos \theta, dX = 3 \cos \theta d\theta \therefore \int \frac{dx}{\sqrt{9-x^2}} = \int \frac{3 \cos \theta d\theta}{3 \cos \theta} = \int 1 d\theta = \theta + C = \sin^{-1} \frac{x}{3} + C$
8. Let  $t = \frac{1}{3} \sin \theta \rightarrow dt = \frac{1}{3} \cos \theta d\theta, \sqrt{1-9t^2} = \cos \theta \therefore \int \sqrt{1-9t^2} dt = \int \frac{1}{3} \cos^2 \theta d\theta = \frac{1}{3} \int \frac{1+\cos 2\theta}{2} d\theta = \frac{1}{6} \left( \theta + \frac{\sin 2\theta}{2} \right) + C = \frac{1}{6} \left( \frac{1}{2} \sin^{-1}(3t) + \frac{6t \sqrt{1-9t^2}}{2} \right) + C$
9. Let  $X = \frac{7}{2} \sec \theta \rightarrow dX = \frac{7}{2} \sec \theta \tan \theta d\theta, \sqrt{4X^2-49} = 7 \tan \theta \therefore \int \frac{dx}{\sqrt{4x^2-49}} = \int \frac{\frac{7}{2} \sec \theta \tan \theta d\theta}{7 \tan \theta} = \int \frac{1}{2} \sec \theta d\theta = \frac{1}{2} \ln |\sec \theta + \tan \theta| + C = \frac{1}{2} \ln \left| \frac{\sqrt{4x^2-49}}{7} + \frac{x}{7} \right| + C$
10. Let  $X = \frac{3}{2} \sec \theta \rightarrow dX = \frac{3}{2} \sec \theta \tan \theta d\theta, \sqrt{25X^2-9} = 3 \tan \theta \therefore \int \frac{5dx}{\sqrt{25x^2-9}} = \int \frac{3 \sec \theta \tan \theta d\theta}{3 \tan \theta} = \int \sec \theta d\theta = \ln |\sec \theta + \tan \theta| + C = \ln \left| \frac{5x}{3} + \frac{\sqrt{25x^2-9}}{3} \right| + C$
15. Let  $X = 3 \sin \theta \rightarrow dX = 3 \cos \theta d\theta, \sqrt{9-X^2} = 3 \cos \theta \left( -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2} \right) \therefore \int \frac{X}{\sqrt{9-X^2}} dX = \int \frac{3 \sin \theta}{3 \cos \theta} (3 \cos \theta d\theta) = \int 3 \sin \theta d\theta = -3 \cos \theta + C = -\sqrt{9-X^2} + C$
17. Let  $X = 2 \tan \theta \rightarrow dX = 2 \sec^2 \theta d\theta, \sqrt{X^2+4} = 2 \sec \theta \therefore \int \frac{X^3 dX}{\sqrt{X^2+4}} = \int \frac{8 \tan^3 \theta (2 \sec^2 \theta d\theta)}{2 \sec \theta} = \int 8 \tan^2 \theta \sec \theta d\theta = \int 8 (\sec^2 \theta - 1) (\sec \theta \tan \theta d\theta) = \int (8 \sec^3 \theta - 8 \sec \theta) d(\sec \theta) = \frac{8}{3} \sec^3 \theta - 8 \sec \theta + C = \frac{(X^2+4)^{3/2}}{3} - 4 \sqrt{X^2+4} + C$
21. Let  $X = \sin \theta \rightarrow dX = \cos \theta d\theta, \sqrt{\frac{X+1}{1-X}} = \sqrt{\frac{1+\sin \theta}{1-\sin \theta}} \cdot \sqrt{\frac{1+\sin \theta}{1+\sin \theta}} = \frac{1+\sin \theta}{\cos \theta} \left( 0 \leq \theta < \frac{\pi}{2} \right) \therefore \int \frac{\sqrt{X+1}}{1-X} dX = \int \frac{(1+\sin \theta) \cos \theta d\theta}{\cos^2 \theta} = \int (1+\sin \theta) d\theta = \theta - \cos \theta + C = \sin^{-1} X - \sqrt{1-X^2} + C$
26. Let  $X = \sec \theta \rightarrow dX = \sec \theta \tan \theta d\theta, (X^2-1) = \tan^2 \theta \therefore \int \frac{X^3 dX}{(X^2-1)^{3/2}} = \int \frac{\sec^3 \theta (\sec \theta \tan \theta d\theta)}{\tan^3 \theta} = \int \frac{\sec^4 \theta}{\tan^3 \theta} d\theta = \int \frac{\sec^4 \theta}{\sin^3 \theta \cos^4 \theta} d\theta = \int \frac{1}{\sin^3 \theta} d(\sin \theta) = -\frac{1}{2 \sin^2 \theta} + C = -\frac{1}{2(1-X^2)} + C = -\frac{X^2}{2(X^2-1)^{3/2}} + C$
36. Let  $t = \ln(\tan \theta) \rightarrow dt = \frac{\sec^2 \theta}{\tan \theta} d\theta, e^t = \tan \theta, e^{2t} + 1 = \sec^2 \theta \therefore \int \frac{e^t dt}{(1+e^{2t})^{3/2}} = \int \frac{\tan \theta d\theta}{\tan^3 \theta (\sec^2 \theta)^{3/2}} = \int \frac{\sec^2 \theta}{\tan^3 \theta \sec^3 \theta} d\theta = \int \frac{1}{\tan^3 \theta} d\theta = \int \frac{\cos^3 \theta}{\sin^3 \theta} d\theta = \int \frac{\cos^2 \theta}{\sin^3 \theta} d\theta = \int \frac{1-\sin^2 \theta}{\sin^3 \theta} d\theta = \int \left( \frac{1}{\sin^3 \theta} - \frac{1}{\sin \theta} \right) d\theta = -\frac{1}{2 \sin^2 \theta} - \ln |\sin \theta| + C = -\frac{1}{2(1-X^2)} - \ln |X| + C$
37. Let  $t = \frac{1}{4} \tan^2 \theta \rightarrow \sqrt{t} = \frac{1}{2} \tan \theta, dt = \frac{1}{2} \tan \theta \cdot \sec^2 \theta d\theta \left( 0 < \theta < \frac{\pi}{2} \right), \sqrt{t} + 4\sqrt{t} = \sqrt{t}(1+4) = \frac{1}{2} \tan \theta (\sec^2 \theta) \therefore \int \frac{dt}{\sqrt{t}+4\sqrt{t}} = \int \frac{\frac{1}{2} \tan \theta \sec^2 \theta d\theta}{\frac{1}{2} \tan \theta (\sec^2 \theta)} = \int \frac{1}{\sec^2 \theta} d\theta = \int \cos^2 \theta d\theta = \frac{1}{2} \left( \theta + \frac{\sin 2\theta}{2} \right) + C = \frac{1}{2} \left( \frac{\pi}{4} + \frac{\sin \frac{\pi}{2}}{2} \right) + C = \frac{\pi}{8} + \frac{1}{4} + C$
57. a)  $dV = X \sqrt{1-X^2} dX \rightarrow V = -\frac{1}{2} \left( \frac{2}{3} (1-X^2)^{3/2} \right) = -\frac{1}{3} (1-X^2)^{3/2}, U = X^2 \rightarrow dU = 2X dX \therefore \int X^3 \sqrt{1-X^2} dX = -\frac{1}{3} X^2 (1-X^2)^{3/2} - \int -\frac{1}{3} (1-X^2)^{3/2} \cdot 2X dX = -\frac{1}{3} X^2 (1-X^2)^{3/2} - \frac{1}{3} \int (1-X^2)^{3/2} \cdot (-2X) dX = -\frac{1}{3} X^2 (1-X^2)^{3/2} - \frac{1}{3} \left( \frac{2}{5} (1-X^2)^{5/2} \right) + C = -\frac{1}{3} X^2 (1-X^2)^{3/2} - \frac{2}{15} (1-X^2)^{5/2} + C$
- b)  $U = 1-X^2 \rightarrow dU = -2X dX \therefore \int X^3 \sqrt{1-X^2} dX = \int (1-U) \sqrt{U} \left( -\frac{1}{2} dU \right) = -\frac{1}{2} \int (U \sqrt{U} - \sqrt{U}) dU = -\frac{1}{2} \left( \frac{2}{5} U^{5/2} - \frac{2}{3} U^{3/2} \right) + C = -\frac{1}{5} (1-X^2)^{5/2} + \frac{1}{3} (1-X^2)^{3/2} + C$
- c) Let  $X = \sin \theta \rightarrow dX = \cos \theta d\theta \therefore \int X^3 \sqrt{1-X^2} dX = \int \sin^3 \theta \cdot \cos \theta \cdot \cos \theta d\theta = \int \sin \theta (1-\cos^2 \theta) (\cos^2 \theta d\theta) = \int (\cos^2 \theta - \cos^4 \theta) (-\sin \theta d\theta) = -\frac{\cos^3 \theta}{3} + \frac{\cos^5 \theta}{5} + C = \frac{(1-X^2)^{3/2}}{3} - \frac{(1-X^2)^{5/2}}{5} + C$

58. a) The slope of line tangent to  $y = f(x)$  is given by  $f'(x)$ , and consider the right triangle with hypotenuse 10, base  $x$  and height  $= \sqrt{100-x^2}$ , we get that  $f'(x) = \frac{h}{x} = -\frac{\sqrt{100-x^2}}{x}$  as the tangent line.

b) Let  $X = 10 \sin \theta \rightarrow dX = 10 \cos \theta d\theta, \sqrt{100-X^2} = 10 \cos \theta \left( 0 < \theta < \frac{\pi}{2} \right)$   
 From  $f(x) = \int f'(x) dx = \int \frac{-\sqrt{100-x^2}}{x} dx = \int \frac{-10 \cos \theta}{10 \sin \theta} (-10 \cos \theta d\theta) = \int \frac{10 \cos^2 \theta}{\sin \theta} d\theta = 10 \int \frac{1-\sin^2 \theta}{\sin \theta} d\theta = 10 \int (\sec \theta - \sin \theta) d\theta = 10 (\ln |\sec \theta + \tan \theta| + \cos \theta) + C = 10 \ln \left| \frac{10 + \sqrt{100-x^2}}{x} \right| + \sqrt{100-x^2} + C$   
 $\therefore f(10) = 0 \therefore 10 \ln \left| \frac{10+0}{10} \right| + 0 + C = 0 \rightarrow C = 0 \therefore f(x) = 10 \ln \left| \frac{10 + \sqrt{100-x^2}}{x} \right| + \sqrt{100-x^2}$

6.  $z^3 - z^2 - 6z = z(z+2)(z-3)$

$$\therefore \frac{z}{z^3 - z^2 - 6z} = \frac{A}{z} + \frac{B}{z+2} + \frac{C}{z-3}$$

$$z = A(z+2)(z-3) + B(z)(z-3) + C(z)(z+2) = A(z^2 - z - 6) + B(z^2 - 3z) + C(z^2 + 2z)$$

Comparing coordinates:  $A+B+C=0$

$$-A-3B+2C=1$$

$$-6A=0 \rightarrow A=0, B=-\frac{1}{3}, C=\frac{1}{3}$$

$$\therefore \frac{z}{z^3 - z^2 - 6z} = -\frac{1}{3(z+2)} + \frac{1}{3(z-3)}$$

7.  $\therefore t^2 - 5t + 6 = (t-2)(t-3)$ , we let  $\frac{t^2+8}{t^2-5t+6} = 1 + \frac{5t+2}{t^2-5t+6}$

and let  $\frac{5t+2}{t^2-5t+6} = \frac{A}{t-2} + \frac{B}{t-3}$

$$5t+2 = A(t-3) + B(t-2)$$

Comparing coordinates:  $A+B=5$

$$-3A-2B=2$$

$$\left. \begin{array}{l} A+B=5 \\ -3A-2B=2 \end{array} \right\} -2A-2B=-10 \rightarrow \begin{array}{l} A=-12 \\ B=17 \end{array}$$

$$\therefore \frac{t^2+8}{t^2-5t+6} = 1 - \frac{12}{t-2} + \frac{17}{t-3}$$

13. Let  $\frac{y}{y^2-2y-3} = \frac{A}{y+1} + \frac{B}{y-3}$

$$y = A(y-3) + B(y+1) = (A+B)y + (-3A+B)$$

$$\therefore A+B=1$$

$$-3A+B=0 \quad \left. \begin{array}{l} A+B=1 \\ -3A+B=0 \end{array} \right\} 4A=1 \therefore A=\frac{1}{4}, B=\frac{3}{4}$$

$$\therefore \int_4^8 \frac{y}{y^2-2y-3} dy = \int_4^8 \left( \frac{1}{4(y+1)} + \frac{3}{4(y-3)} \right) dy = \left[ \frac{1}{4} \ln|y+1| + \frac{3}{4} \ln|y-3| \right]_4^8 = \frac{1}{4}(\ln 9 - \ln 5) + \frac{3}{4} \ln 5 = \frac{1}{2}(\ln 3 + \ln 5)$$

16. Let  $\frac{x+3}{2x^2-8x} = \frac{A}{x} + \frac{B}{x-2} + \frac{C}{x+2}$

$$x+3 = 2A(x^2-4) + 2B(x^2+2x) + 2C(x^2-2x) = (2A+2B+2C)x^2 + (4B-4C)x + (-8A)$$

Comparing coordinates:  $2A+2B+2C=0$

$$4B-4C=1$$

$$-8A=3 \rightarrow A=-\frac{3}{8} \therefore B+C=\frac{3}{8}, B-C=\frac{1}{4} \therefore B=\frac{5}{16}, C=\frac{1}{16}$$

$$\therefore \int \frac{x+3}{2x^2-8x} dx = \int \left( -\frac{3}{8x} + \frac{5}{16(x-2)} + \frac{1}{16(x+2)} \right) dx$$

$$= -\frac{3}{8} \ln|x| + \frac{5}{16} \ln|x-2| + \frac{1}{16} \ln|x+2| + C$$

20. Let  $\frac{x^2}{(x-1)(x^2+2x+1)} = \frac{A}{x-1} + \frac{B}{x+1} + \frac{C}{(x+1)^2}$

$$x^2 = A(x^2+2x+1) + B(x^2-1) + C(x-1) = (A+B)x^2 + (2A+C)x + (A-B-C)$$

Comparing coordinates:  $A+B=1$  ①

$$①-③: 2A-C=1$$

$$2A+C=0$$
 ②

$$\therefore A=\frac{1}{4}, C=-\frac{1}{2} \rightarrow B=\frac{3}{4}$$

$$A-B-C=0$$
 ③

$$\therefore \int \frac{x^2}{(x-1)(x^2+2x+1)} dx = \int \left( \frac{1}{4(x-1)} + \frac{3}{4(x+1)} - \frac{1}{2(x+1)^2} \right) dx = \frac{1}{4} \ln|x-1| + \frac{3}{4} \ln|x+1| + \frac{1}{2(x+1)} + C$$

23.  $\int \frac{y^2+2y+1}{(y^2+1)^2} dy = \int \frac{y^2+1}{(y^2+1)^2} dy + \int \frac{2y}{(y^2+1)^2} dy = \int \frac{1}{y^2+1} dy + \int \frac{1}{u} du$  (Let  $u=y^2+1 \rightarrow du=2y dy$ )

$$= \arctan(y) - \frac{1}{y^2+1} + C$$

27.  $\frac{x^2-x+2}{x^2-1} = \frac{A}{x-1} + \frac{Bx+C}{x^2-x+1}$

$\therefore x^2-x+2 = A(x^2+x+1) + (Bx+C)(x-1) = (A+B)x^2 + (A+C-B)x + (A-C)$

$\therefore A+B=1 \rightarrow A=\frac{2}{3}, C=-\frac{4}{3}, B=\frac{1}{3}$

$A+C-B=-1 \therefore \int \frac{x^2-x+2}{x^2-1} dx = \int \left( \frac{2}{3(x-1)} + \frac{x-4}{3(x^2-x+1)} \right) dx$   
 $= \frac{2}{3} \ln|x-1| + \int \frac{x-4}{3(x^2-x+1)} dx$   
 $= \frac{2}{3} \ln|x-1| + \frac{1}{3} \int \frac{x-4}{(x-\frac{1}{2})^2 + \frac{3}{4}} dx$   
 $= \frac{2}{3} \ln|x-1| + \frac{1}{3} \int \frac{u-\frac{7}{2}}{u^2 + \frac{3}{4}} du \quad (u=x-\frac{1}{2})$   
 $= \frac{2}{3} \ln|x-1| + \frac{1}{6} \ln|u^2 + \frac{3}{4}| - \int \frac{\frac{7}{2}}{2(u^2 + \frac{3}{4})} du$   
 $= \frac{2}{3} \ln|x-1| + \frac{1}{6} \ln(u^2 + \frac{3}{4}) - \frac{3}{2} \int \frac{\frac{1}{2}}{(u^2 + \frac{3}{4})} du \rightarrow 2 \cdot \frac{\sqrt{3}}{2} \arctan(\frac{2}{\sqrt{3}}u)$   
 $= \frac{2}{3} \ln|x-1| + \frac{1}{6} \ln(x^2-x+1) - \sqrt{3} \arctan(\frac{2x+1}{\sqrt{3}}) + C$

31.  $\frac{2\theta^3+5\theta^2+8\theta+4}{(\theta^2+2\theta+2)^2} = \frac{A\theta+B}{\theta^2+2\theta+2} + \frac{C\theta+D}{(\theta^2+2\theta+2)^2}$

$2\theta^3+5\theta^2+8\theta+4 = (A\theta+B)(\theta^2+2\theta+2) + C\theta+D = A\theta^3 + (B+2A)\theta^2 + (C+2B+2A)\theta + (2B+D)$

$\rightarrow A=2, B=1, C=2, D=2$

$\therefore \int \frac{2\theta^3+5\theta^2+8\theta+4}{(\theta^2+2\theta+2)^2} d\theta = \int \left( \frac{2\theta+1}{\theta^2+2\theta+2} + \frac{2\theta+2}{(\theta^2+2\theta+2)^2} \right) d\theta$   
 $\stackrel{\frac{1}{u} du}{=} \int \frac{2\theta+2}{\theta^2+2\theta+2} d\theta - \int \frac{1}{(\theta^2+1)^2} d\theta + \int \frac{1}{u^2} du \quad (u=\theta^2+2\theta+2 \rightarrow du=(2\theta+2)d\theta)$   
 $= \ln|\theta^2+2\theta+2| - \arctan(\theta+1) - \frac{1}{\theta^2+2\theta+2} + C$

34.  $\int \frac{x^4}{x^2-1} dx = \int (x^2+1) + \frac{1}{x^2-1} dx = \int (x^2+1) dx + \int \frac{1}{x^2-1} dx$   
 $= \frac{x^3}{3} + x + C + \left( \int \frac{1}{\tan^2 \theta} \cdot \sec \theta \tan \theta d\theta \right) \quad x = \sec \theta$   
 $= \frac{x^3}{3} + x + C + \left( \int \csc \theta d\theta \right) = \frac{x^3}{3} + x - \ln|\csc x + \cot x| + C$   
 $= \frac{x^3}{3} + x - \ln \left| \frac{x+1}{x^2-1} \right| + C$

35.  $\int \frac{9x^2-3x+1}{x^2-x^2} dx = \int 9 + \frac{9x^2-3x+1}{x^2-x^2} dx$   
 $= 9x + \int \left( \frac{7}{x-1} + \frac{2}{x} - \frac{1}{x} \right) dx$   
 $= 9x + 7 \ln|x-1| + 2 \ln|x| + \frac{1}{x} + C$

$\frac{9x^2-3x+1}{x^2-x^2} = \frac{A}{x-1} + \frac{B}{x} + \frac{C}{x}$

$9x^2-3x+1 = Ax^2 + Bx^2 + Cx - C$

$A+B=9, -B+C=-3, -C=1$

$\rightarrow C=-1, B=2, A=7$

41.  $\int \frac{\cos y dy}{\sin^2 y + \sin y - 6} = \int \frac{1}{u^2+u-6} du = \int \frac{1}{5(u-\frac{1}{2}) - \frac{1}{2}} du = \frac{1}{5} \ln|u-2| + \frac{1}{5} \ln|u-3| + C$   
 $(u=\sin y \rightarrow du=\cos y dy)$   
 $= \frac{1}{5} \ln(2-\sin y) + \frac{1}{5} \ln(3-\sin y) + C$

## Section 8.7

8.  $\int_2^4 \frac{1}{s-1} ds \quad f'''(x) = \frac{6}{(s-1)^4} \quad f^{(4)}(x) = \frac{120}{(s-1)^5}$

I. a)  $y_0=1, y_1=\frac{4}{9}, y_2=\frac{1}{4}, y_3=\frac{4}{25}, y_4=\frac{1}{9}$

$T = \frac{1/2}{2} \left( 1 + \frac{8}{9} + \frac{1}{2} + \frac{8}{25} + \frac{1}{9} \right) = 0.705$

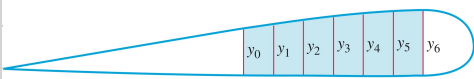
$|E_T| \leq \frac{b-a}{12} \max |f^{(4)}(x)| (\Delta x)^2 = \frac{2}{12} \left| \frac{6}{(2-1)^4} \right| \left( \frac{1}{2} \right)^2 = 0.25$

b)  $\int_2^4 \frac{1}{(s-1)^2} ds = \left[ -\frac{1}{s-1} \right]_2^4 = \frac{2}{3} \therefore |E_T| = |0.667 - 0.705| = 0.038$

c)  $\frac{|E_T|}{\text{True Value}} \times 100 = \frac{0.038}{0.667} \times 100 = 5.7\%$

8. II. a)  $S = \frac{112}{3} \left( 1 + \frac{16}{9} + \frac{1}{2} + \frac{16}{25} + \frac{1}{9} \right) = 0.6715$   
 $|E_s| \leq \frac{b-a}{180} \max |f^{(4)}(x)| (\Delta x)^4 = \frac{2}{180} \left( \frac{120}{(2-0)^4} \right) \left( \frac{1}{2} \right)^4 = 0.0833$   
 b)  $\int_2^9 \frac{1}{\sqrt{s}} ds = \frac{2}{3} = 0.6667 \rightarrow |E_s| = |0.6667 - 0.6715| = 0.0048$   
 c) % error =  $\frac{|E_s|}{\text{True value}} = \frac{0.0048}{0.6667} \times 100 = 0.72\%$

24. Distance =  $\sum_{i=0}^{10} \frac{1}{2} (y_{i+1} + y_i) \Delta x_i$   $y_i$  = Speed at time  $x_i$   
 $\Delta x_i = x_{i+1} - x_i \quad i = 0, 1, \dots, 10$   
 $= \frac{1}{2}(30)(2.2) + \frac{1}{2}(70)(1.0) + \frac{1}{2}(90)(1.3) + \frac{1}{2}(110)(1.4) + \frac{1}{2}(130)(1.9) + \frac{1}{2}(150)(2.4) + \frac{1}{2}(170)(2.5) + \frac{1}{2}(190)(3.3) + \frac{1}{2}(210)(4.6)$   
 $+ \frac{1}{2}(230)(5.6) + \frac{1}{2}(250)(10.9)$   
 $= 3522.5 \text{ km/h} \cdot \text{sec} \cdot 1\text{h}/3600\text{sec} = 0.978 \text{ km}$

25.   
 $y_0 = 0.5 \text{ m}, y_1 = 0.55 \text{ m}, y_2 = 0.6 \text{ m}, y_3 = 0.65 \text{ m}, y_4 = 0.7 \text{ m}, y_5 = 0.75 \text{ m}, y_6 = 0.75 \text{ m}$  Horizontal spacing = 0.3 m  
 Cross section area:  
 $S = \frac{0.3}{3} (0.50 + 2.90 + 1.20 + 2.60 + 1.40 + 3.00 + 0.75) = 1.165 \text{ m}^2$   
 Gasoline volume =  $\frac{2000 \text{ kg}}{673 \text{ kg/m}^3} = 2.9718 \text{ m}^3$   
 $\therefore \text{Tank length} = \frac{2.9718 \text{ m}^3}{1.165 \text{ m}^2} = 2.55 \text{ m}$

28. a)  $\Delta x = \frac{1}{10} = 0.1 \quad k_0 = e^{-0^2} = 1, k_1 = e^{-0.1^2} = 0.99, k_2 = 0.96, k_3 = 0.91, k_4 = 0.85, k_5 = 0.78, k_6 = 0.70$   
 $k_7 = 0.61, k_8 = 0.53, k_9 = 0.44, k_{10} = 0.37 \quad (y_i = \sum_{k=0}^i k_k; i = 0, 1, \dots, 9)$

Using Simpson's rule, we get

erf(1) =  $S = \frac{0.1}{3} \left( \frac{2}{10} \right) (1 + 3.96 + 1.92 + 3.64 + 1.70 + 3.12 + 1.40 + 2.44 + 1.06 + 1.76 + 0.37) = 0.8414$

b)  $|E_s| \leq \frac{b-a}{180} \max |f^{(4)}(x)| (\Delta x)^4 = \frac{1}{180} (12) \left( \frac{1}{10} \right)^4 = 6.67 \times 10^{-6}$

29. Sum  $T$  in trapezoidal rule of  $\int_a^b f(x) dx \rightarrow T = \sum_{k=0}^{n-1} \frac{1}{2} (f(x_{k+1}) + f(x_k)) \Delta x$

Since  $\frac{1}{2}(f(x_{k+1}) + f(x_k))$  is between  $f(x_{k+1})$  and  $f(x_k)$ , by IVT, there exists  $c_k$  such that

$f(c_k) = \frac{1}{2}(f(x_{k+1}) + f(x_k))$ , and if we choose such  $c_k$ , we get  $T = \sum_{k=0}^{n-1} (f(c_k) \Delta x)$ , which is also the Riemann sum for  $f$  continuous on  $[a, b]$ .

31. a) Length =  $4 \int_0^{\pi/2} \sqrt{1 - \frac{1}{4} \cos^2 t} dt$

$\therefore y_k = 4 \sqrt{1 - \frac{1}{4} \cos^2 x_k} \quad ; \quad \Delta x = \frac{\pi/2}{10} = \frac{\pi}{20}, \quad x_k = \frac{\pi}{20} \cdot k$

$y_0 = 2.828, y_1 = 3.478, y_2 = 3.519, y_3 = 3.581, y_4 = 3.658, y_5 = 3.742, y_6 = 3.823, y_7 = 3.896,$

$y_8 = 3.952, y_9 = 3.988, y_{10} = 4$

$\therefore T = \frac{\pi/20}{2} (2.828 + 6.956 + 7.038 + 7.162 + 7.316 + 7.484 + 7.646 + 7.792 + 7.904 + 7.976 + 4.000)$   
 $= 5.805$

b)  $\therefore \max |f''(t)| = \max \left| \frac{d^2}{dt^2} (\sqrt{1 - e^2 \cos^2 t}) \right| \leq 1$

$\therefore |E_T| \leq \frac{b-a}{12} \max |f''(x)| (\Delta x)^2 = \frac{\pi}{24} (1) \left( \frac{\pi}{20} \right)^2 = 0.00323$